

Problem Statement

Understanding customer preferences and restaurant trends is important for making informed business decisions in food industry. In this article, we will analyze Zomato's restaurant dataset using Python to find meaningful insights. We aim to answer questions such as:

Do more restaurants provide online delivery compared to offline services?

Which types of restaurants are most favored by the general public?

What price range do couples prefer for dining out?

Step 1: Importing necessary Python libraries. We will be using Pandas, Numpy, Matplotlib and Seaborn libraries.

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Creating the data frame.

```
In [4]: dataframe = pd.read_csv("Zomato-data.csv")
print(dataframe.head())
```

	name	online_order	book_table	rate	votes	\
0	Jalsa	Yes	Yes	4.1/5	775	
1	Spice Elephant	Yes	No	4.1/5	787	
2	San Churro Cafe	Yes	No	3.8/5	918	
3	Addhuri Udipi Bhojana	No	No	3.7/5	88	
4	Grand Village	No	No	3.8/5	166	

	approx_cost(for two people)	listed_in(type)
0	800	Buffet
1	800	Buffet
2	800	Buffet
3	300	Buffet
4	600	Buffet

Step 3: Data Cleaning and Preparation Before moving further we need to clean and process the data.

1. Convert the rate column to a float by removing denominator characters.

`dataframe['rate']=dataframe['rate'].apply(handleRate):` Applies the `handleRate` function to clean and convert each rating value in the 'rate' column.

```
In [5]: def handleRate(value):
value=str(value).split('/')
value=value[0];
return float(value)
```

```
dataframe['rate']=dataframe['rate'].apply(handleRate)
print(dataframe.head())
```

	name	online_order	book_table	rate	votes	\
0	Jalsa	Yes	Yes	4.1	775	
1	Spice Elephant	Yes	No	4.1	787	
2	San Churro Cafe	Yes	No	3.8	918	
3	Addhuri Udipi Bhojana	No	No	3.7	88	
4	Grand Village	No	No	3.8	166	

	approx_cost(for two people)	listed_in(type)
0	800	Buffet
1	800	Buffet
2	800	Buffet
3	300	Buffet
4	600	Buffet

2. Getting summary of the dataframe use df.info().

```
In [6]: dataframe.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                  148 non-null    object
1   online_order                         148 non-null    object
2   book_table                           148 non-null    object
3   rate                                 148 non-null    float64
4   votes                                148 non-null    int64
5   approx_cost(for two people)          148 non-null    int64
6   listed_in(type)                      148 non-null    object
dtypes: float64(1), int64(2), object(4)
memory usage: 8.2+ KB
```

3. Checking for missing or null values to identify any data gaps.

```
In [7]: print(dataframe.isnull().sum())
```

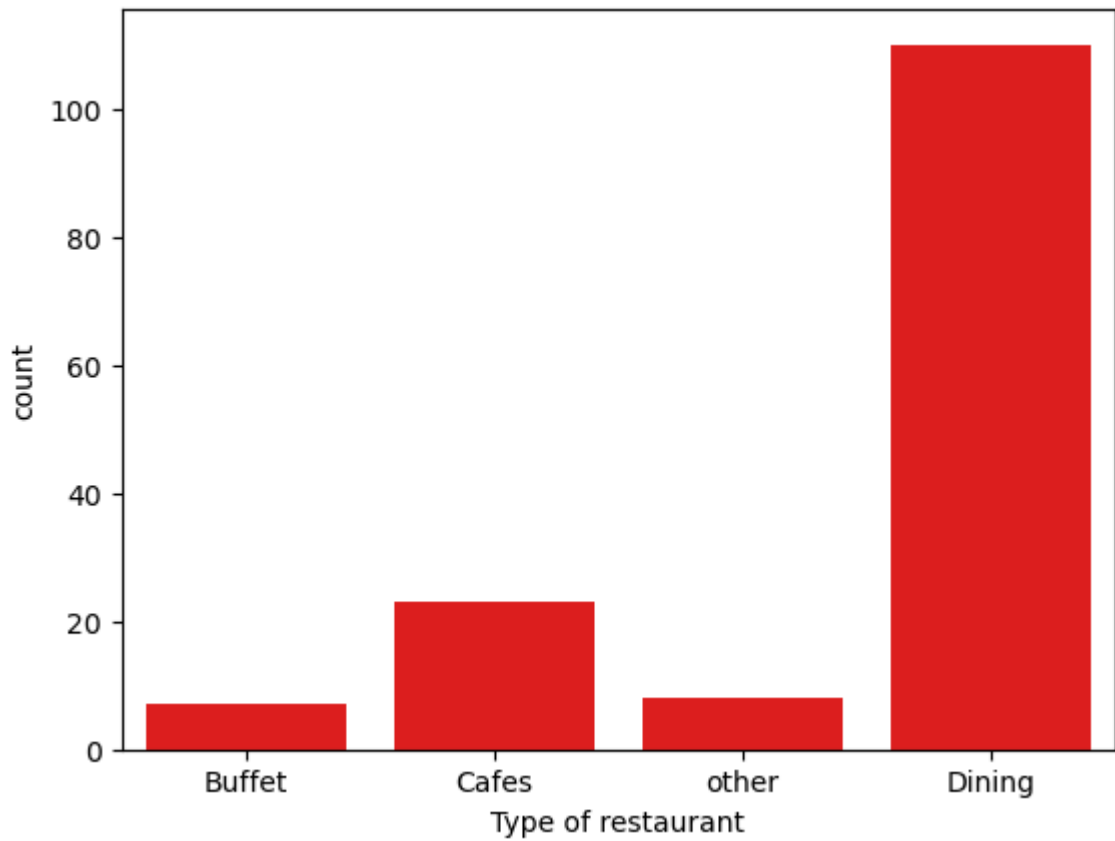
```
name                                0
online_order                        0
book_table                          0
rate                                0
votes                               0
approx_cost(for two people)         0
listed_in(type)                     0
dtype: int64
```

Step 4: Exploring Restaurant Types

1. Let's see the listed_in (type) column to identify popular restaurant categories.

```
In [8]: sns.countplot(x=dataframe['listed_in(type)'],color='red')
plt.xlabel("Type of restaurant")
```

```
Out[8]: Text(0.5, 0, 'Type of restaurant')
```

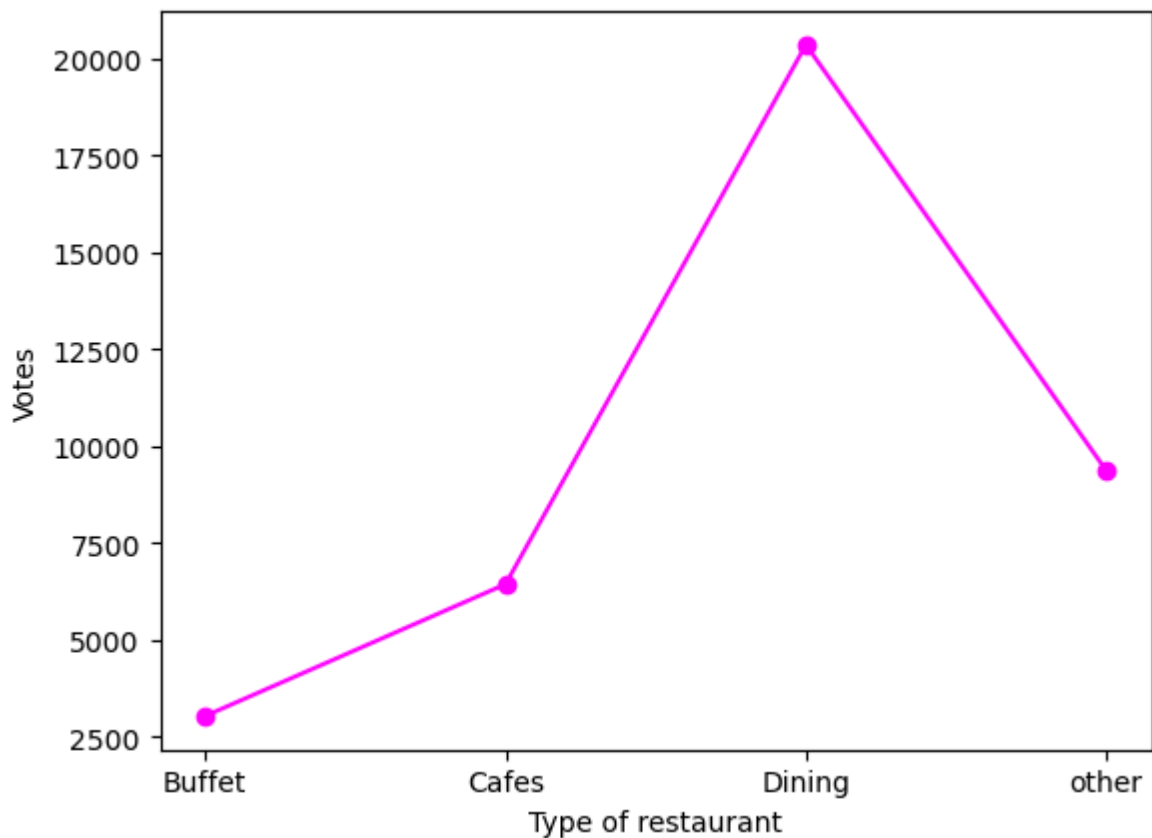


Conclusion: The majority of the restaurants fall into the dining category.

2. Votes by Restaurant Type

```
In [13]: grouped_data = dataframe.groupby('listed_in(type)')['votes'].sum()
result = pd.DataFrame({'votes': grouped_data})
plt.plot(result, c='magenta', marker='o')
plt.xlabel('Type of restaurant')
plt.ylabel('Votes')
```

Out[13]: Text(0, 0.5, 'Votes')



Conclusion: Dining restaurants are preferred by a larger number of individuals.

Step 5: Identify the Most Voted Restaurant

Find the restaurant with the highest number of votes.

```
In [14]: max_votes = dataframe['votes'].max()
restaurant_with_max_votes = dataframe.loc[dataframe['votes'] == max_votes, 'name']

print('Restaurant(s) with the maximum votes:')
print(restaurant_with_max_votes)
```

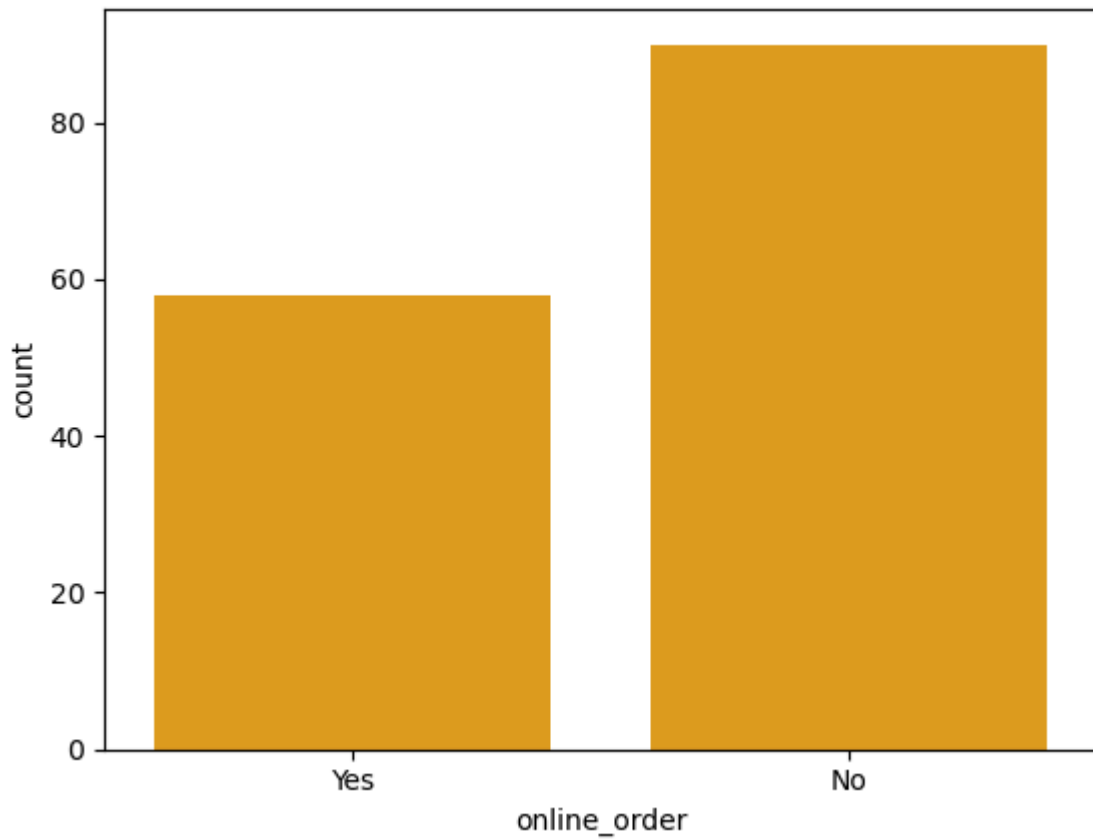
```
Restaurant(s) with the maximum votes:
38    Empire Restaurant
Name: name, dtype: object
```

Step 6: Online Order Availability

Exploring the online_order column to see how many restaurants accept online orders.

```
In [15]: sns.countplot(x=dataframe['online_order'], color='orange')
```

```
Out[15]: <Axes: xlabel='online_order', ylabel='count'>
```

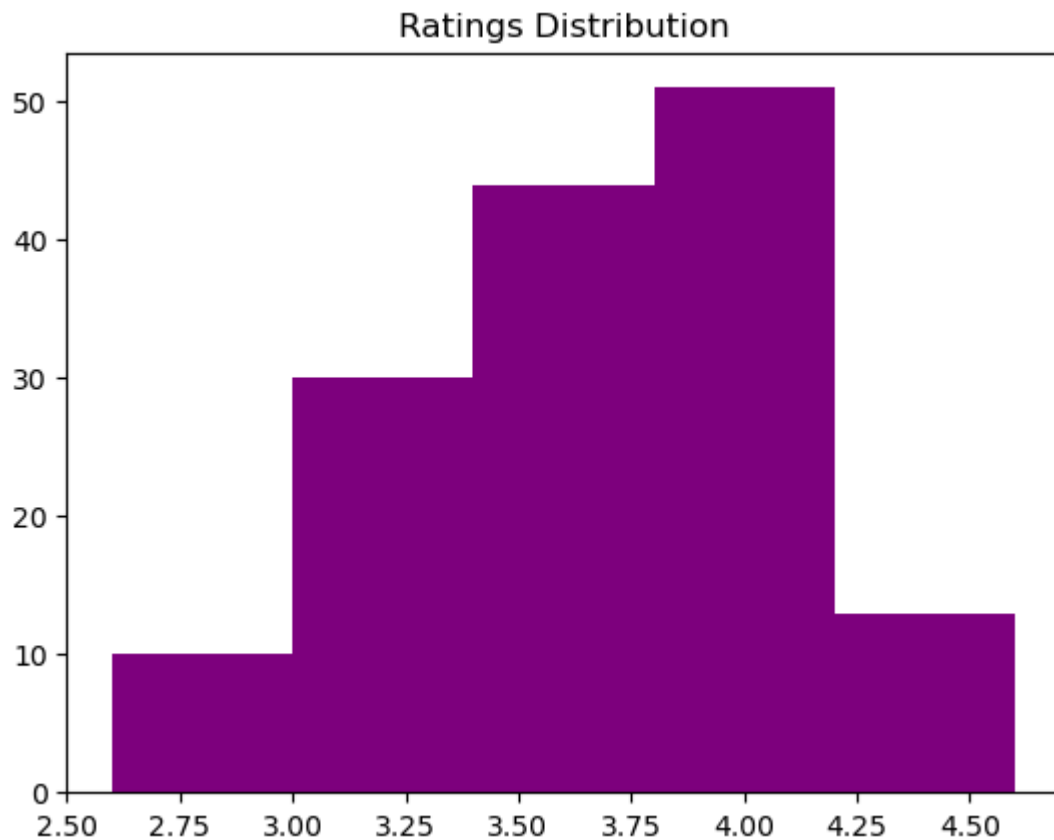


Conclusion: This suggests that a majority of the restaurants do not accept online orders.

Step 7: Analyze Ratings

Checking the distribution of ratings from the rate column.

```
In [16]: plt.hist(dataframe['rate'],bins=5,color='purple')
plt.title('Ratings Distribution')
plt.show()
```



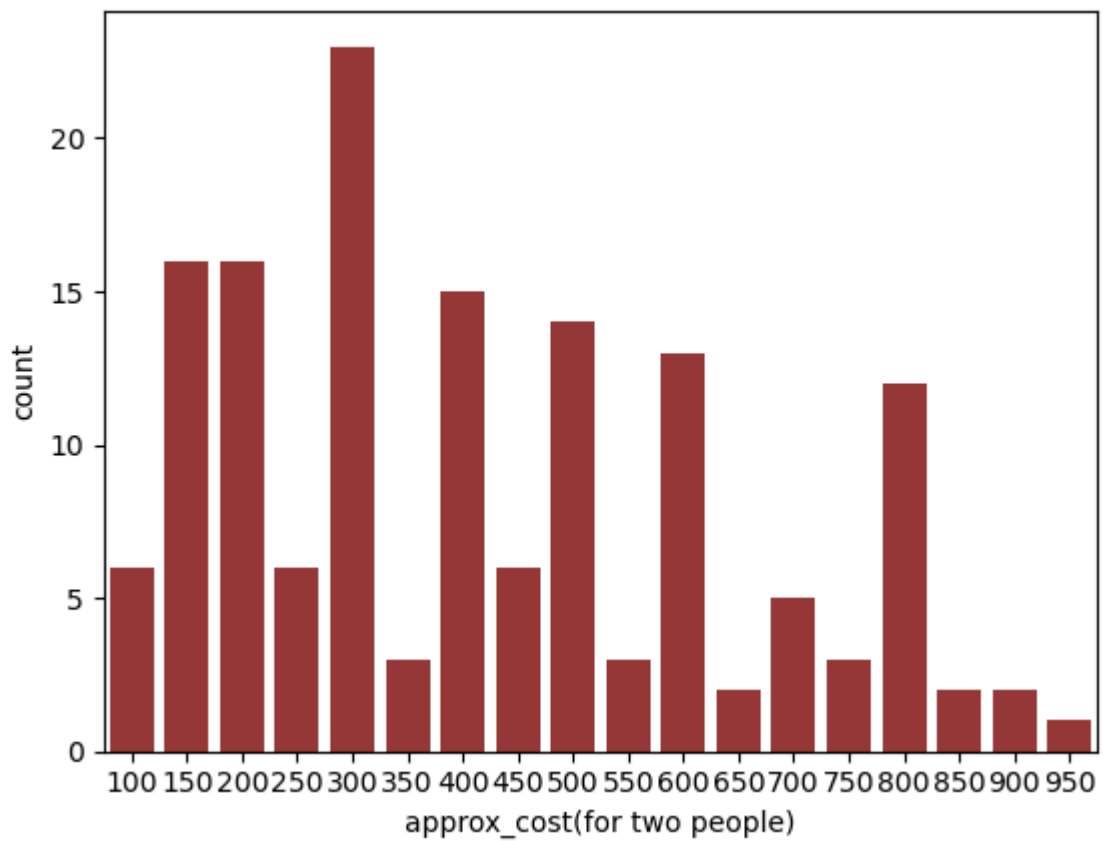
Conclusion: The majority of restaurants received ratings ranging from 3.5 to 4.

Step 8: Approximate Cost for Couples

Analyze the `approx_cost(for two people)` column to find the preferred price range.

```
In [20]: couple_data=dataframe['approx_cost(for two people)']  
sns.countplot(x=couple_data,color='brown')
```

```
Out[20]: <Axes: xlabel='approx_cost(for two people)', ylabel='count'>
```



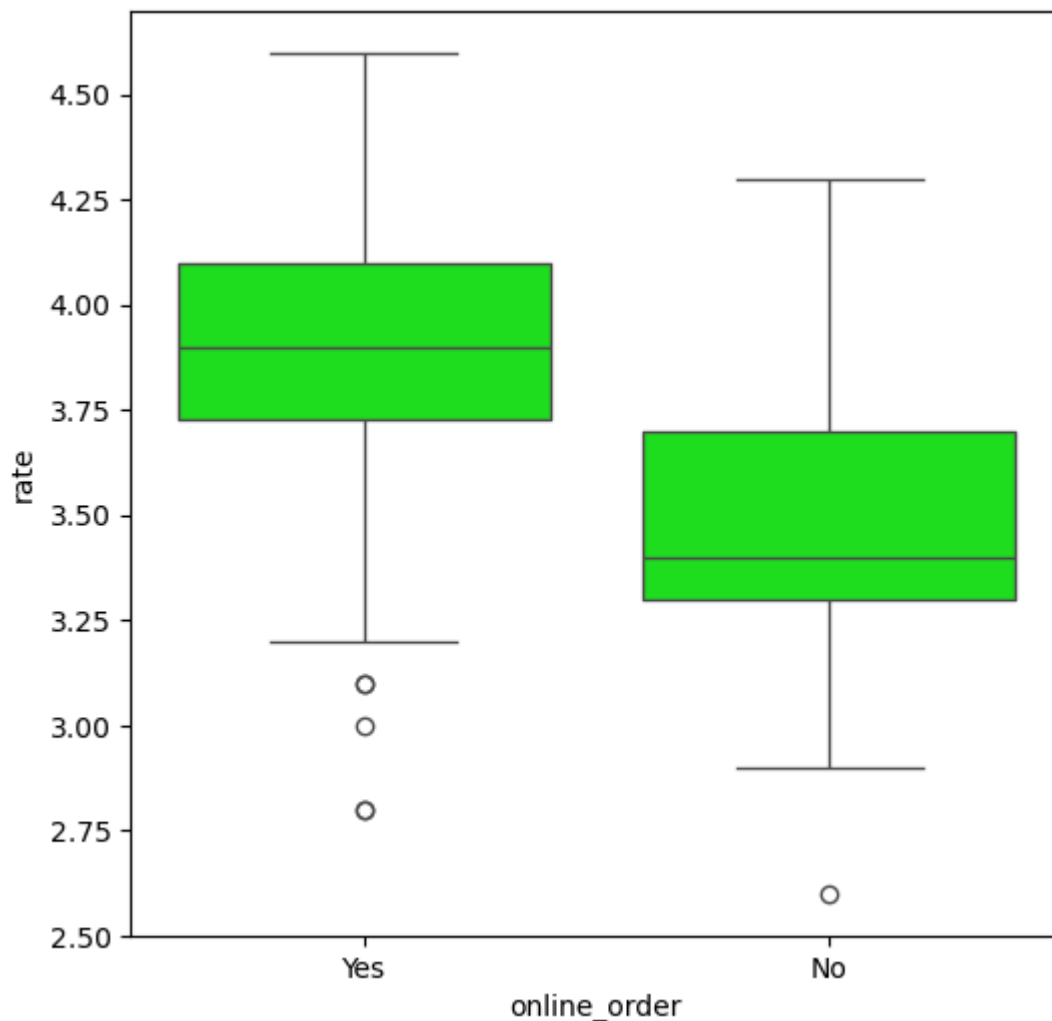
Conclusion: The majority of couples prefer restaurants with an approximate cost of 300 rupees.

Step 9: Ratings Comparison - Online vs Offline Orders

Compare ratings between restaurants that accept online orders and those that don't.

```
In [21]: plt.figure(figsize = (6,6))
sns.boxplot(x = 'online_order', y = 'rate', data = dataframe,color='lime')
```

```
Out[21]: <Axes: xlabel='online_order', ylabel='rate'>
```



Conclusion: Offline orders received lower ratings in comparison to online orders which obtained excellent ratings.

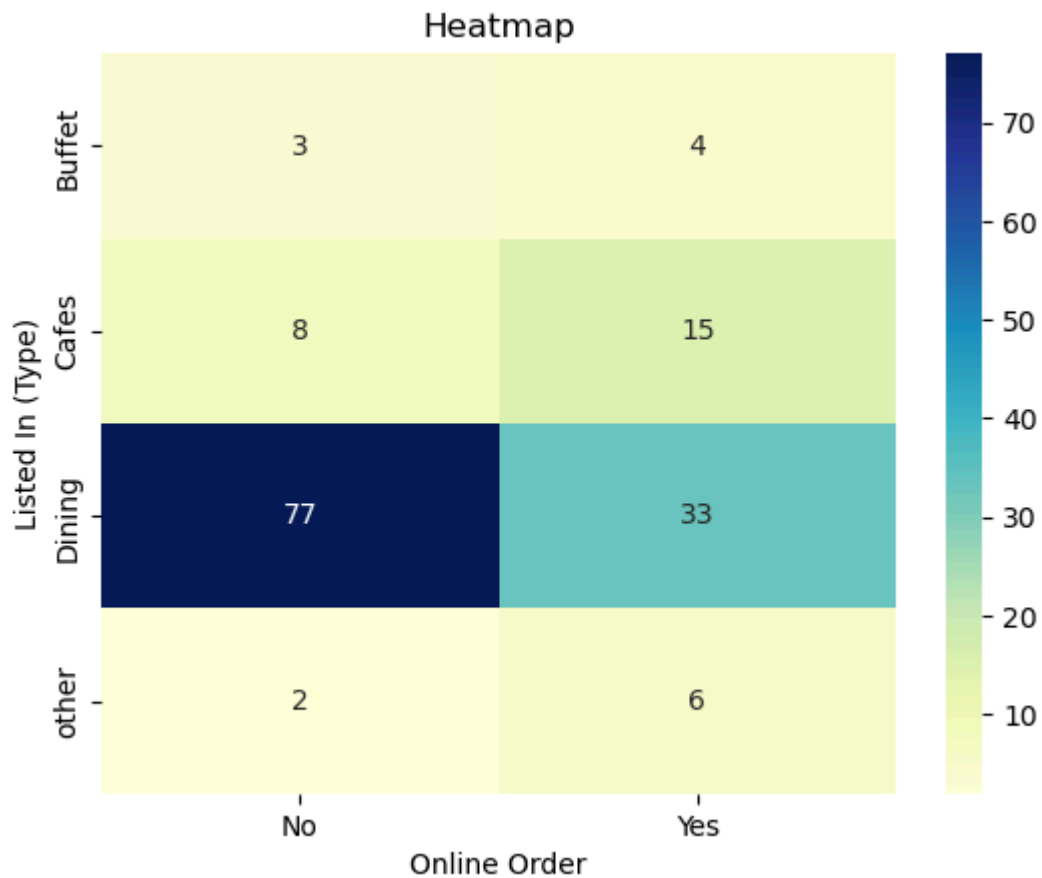
Step 10: Order Mode Preferences by Restaurant Type

Find the relationship between order mode (online_order) and restaurant type (listed_in(type)).

```
pivot_table = dataframe.pivot_table(index='listed_in(type)', columns='online_order',
aggfunc='size', fill_value=0):
```

Creates a pivot table counting restaurants by type and online order availability.

```
In [18]: pivot_table = dataframe.pivot_table(index='listed_in(type)', columns='online_order',
sns.heatmap(pivot_table, annot=True, cmap='YlGnBu', fmt='d')
plt.title('Heatmap')
plt.xlabel('Online Order')
plt.ylabel('Listed In (Type)')
plt.show()
```

With this we can say that dining restaurants primarily accept offline orders whereas cafes primarily receive online orders. This suggests that clients prefer to place orders in person at restaurants but prefer online ordering at cafes.